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SkillGrad: Optimizing Agent Skills Like Gradient Descent

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SkillGrad：Motivation

Motivation：skill能提供领域工作流，但是skill的质量很关键

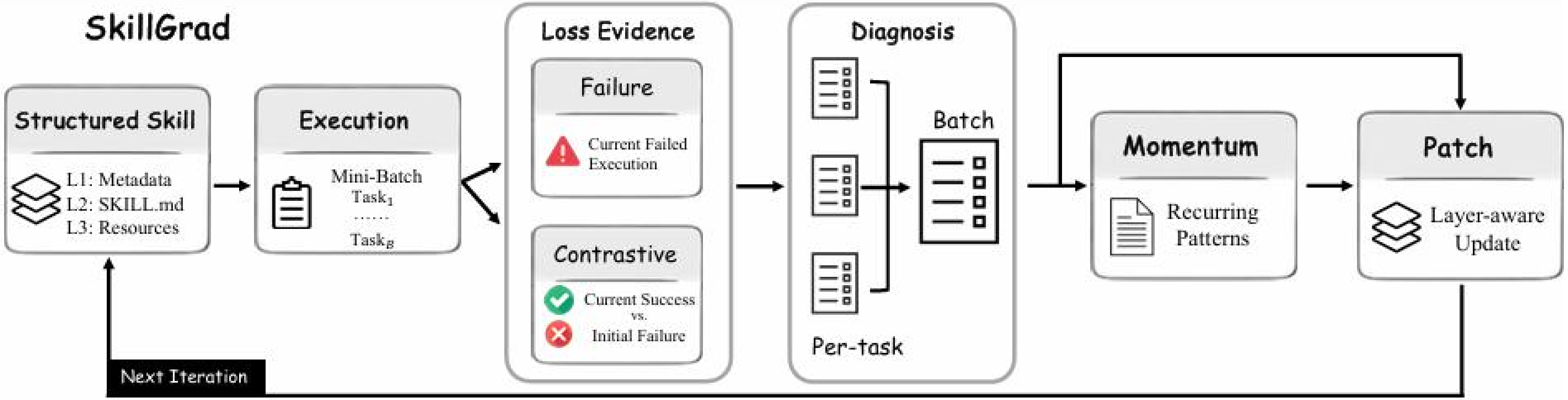
LLM自动生成的/第三方下载的：可能不完整，没有覆盖边缘情况等

能不能把skill当成一个可优化对象？已有方法让模型反思失败并修改skill，但是偏启发式

把 Agent Skill 文件包当作可优化参数

定义loss evidence, gradient-like signal, momentum, update...

本质上可以看成mutli-agent system：LLM diagnoser，momentum，patcher...



Methods

Parameter $S = (H, B, Q),$

- H： metadata header
- B： skill.md， always loaded
- Q： conditionally loaded resources

优化skill不仅要决定加什么知识，还要决定放在哪一层（通用？复杂细节？）

Placing narrow task detail in B can distract the executor on future tasks, while placing core work flow guidance only in Q can prevent the executor from loading it when needed.

Methods

Parameter： $S=(H,B,Q),$

Loss Evidence： $\mathcal{R}(S;\mathcal{T})=\begin{cases}1, & \text{if the agent output passes,}\\ 0, & \text{otherwise.}\end{cases}$ $\mathcal{L}_{\text{bin}}(S;\mathcal{T})=1-\mathcal{R}(S;\mathcal{T}).$

但是binary信号太稀疏：成功的任务也有学习价值，可能包含了一个值得保留的策略

$$\mathcal{E}_t(\mathcal{T})=\begin{cases}\{\tau_t^-,c_t\}, & r_t(\mathcal{T})=0,\\ \{\tau_t^+,\tau_0^-,c_0\}, & r_t(\mathcal{T})=1,\end{cases} \quad (1)$$

c_t： iteration t 执行后的evaluator feedback

τ_t^-,τ_t^+： 使用当前S_t， iteration t 执行，成功或失败的轨迹

如果失败，就用当前失败轨迹和 evaluator feedback 找错误原因；

如果成功，但初始失败，就对比当前成功轨迹和初始失败轨迹，找出成功行为并保留。

Methods

Parameter： $S = (H, B, Q),$

Loss Evidence： $\mathcal{R}(S; \mathcal{T}) = \begin{cases} 1, & \text{if the agent output passes,} \\ 0, & \text{otherwise.} \end{cases}$ $\mathcal{L}_{\text{bin}}(S; \mathcal{T}) = 1 - \mathcal{R}(S; \mathcal{T}).$

$$\mathcal{E}_t(\mathcal{T}) = \begin{cases} \{\tau_t^-, c_t\}, & r_t(\mathcal{T}) = 0, \\ \{\tau_t^+, \tau_0^-, c_0\}, & r_t(\mathcal{T}) = 1, \end{cases} \quad (1)$$

Gradient Signals：类比传统梯度下降，为当前skill生成诊断

$$g_t = \frac{1}{|\mathcal{B}|} \sum_{x_i \in \mathcal{B}} \nabla_\theta \mathcal{L}(\theta_t; x_i). \qquad d_{t,i} = \mathrm{Diag}(S_t, \mathcal{T}_{t,i}, e_{t,i}). \qquad D_t = \{d_{t,1}, d_{t,2}, \ldots, d_{t,B}\}.$$

每个诊断需要说明：如果失败，为什么输出和gt不同，一般什么行为可以避免？如果成功，相对初始失败改变了什么。

Methods

Parameter： $S = (H, B, Q),$

Loss Evidence： $\mathcal{R}(S;\mathcal{T}) = \begin{cases} 1, & \text{if the agent output passes,} \\ 0, & \text{otherwise.} \end{cases}$ $\mathcal{L}_{\text{bin}}(S;\mathcal{T}) = 1 - \mathcal{R}(S;\mathcal{T}).$

$$\mathcal{E}_t(\mathcal{T}) = \begin{cases} \{\tau_t^-, c_t\}, & r_t(\mathcal{T}) = 0, \\ \{\tau_t^+, \tau_0^-, c_0\}, & r_t(\mathcal{T}) = 1, \end{cases} \quad (1)$$

Gradient Signals： $d_{t,i} = \text{Diag}(S_t, \mathcal{T}_{t,i}, e_{t,i}).$ $D_t = \{d_{t,1}, d_{t,2}, \ldots, d_{t,B}\}.$

Momentum：文本记忆稳定优化方向

$v_t = \gamma v_{t-1} + g_t,$ $M_t, O_t = \text{Momentum}(M_{t-1}, D_t, S_t),$

M_t记录cross-iteration patterns，O_t是compact overlay for current patch

Semantic accumulation + conditional update + success preserve

Methods

Parameter： $S = (H, B, Q),$

Loss Evidence： $\mathcal{R}(S;\mathcal{T}) = \begin{cases} 1, & \text{if the agent output passes,} \\ 0, & \text{otherwise.} \end{cases}$ $\mathcal{L}_{\text{bin}}(S;\mathcal{T}) = 1 - \mathcal{R}(S;\mathcal{T}).$

$$\mathcal{E}_t(\mathcal{T}) = \begin{cases} \{\tau_t^-, c_t\}, & r_t(\mathcal{T}) = 0, \\ \{\tau_t^+, \tau_0^-, c_0\}, & r_t(\mathcal{T}) = 1, \end{cases} \qquad (1)$$

Gradient Signals： $d_{t,i} = \text{Diag}(S_t, \mathcal{T}_{t,i}, e_{t,i}).$ $D_t = \{d_{t,1}, d_{t,2}, \ldots, d_{t,B}\}.$

Momentum：文本记忆稳定优化方向 $M_t, O_t = \text{Momentum}(M_{t-1}, D_t, S_t),$

Patcher：分层更新skill $\theta_{t+1} = \theta_t - \eta v_t.$ $S_{t+1} = \text{Patch}(S_t, D_t, M_t, O_t).$

多个任务指向同一机制的话，就只写一个泛化规则；

需要决定更新metadata，SKILL.md还是resource文件

Experiments

SpreadsheetBench Verified，WikiTableQuestions，no model mixture

SkillGrad优于training-free和training-based的baseline

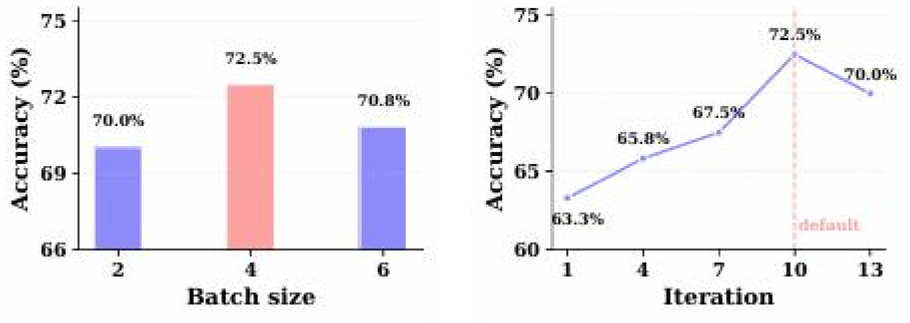
尽管LLM-gen skills会导致模型性能下降，SkillGrad可以弥补这一点

Init.	Method	GPT-5.4		GPT-4.1	
		SpreadsheetBench	**WikiTQ (OOD)**	**SpreadsheetBench**	**WikiTQ (OOD)**
Training-free					
–	No Skill	62.50	78.57	44.17	52.86
LLM-gen.	Base xlsx Skill	55.83	77.14	36.67	48.58
Third-party	Base xlsx Skill	60.00	78.57	33.33	42.86
Training-based skill improvement					
LLM-gen.	Trace2Skill	65.28 ± 2.93	79.05 ± 1.35	37.22 ± 3.54	60.00 ± 5.83
	EvoSkill	68.06 ± 0.48	78.09 ± 0.67	37.22 ± 2.58	53.33 ± 7.50
	SkillGrad (Ours)	**71.11** ± 1.73	**82.38** ± 1.78	**54.17** ± 3.54	**73.65** ± 2.50
Third-party	Trace2Skill	63.89 ± 0.48	81.91 ± 1.78	38.89 ± 3.85	51.43 ± 1.17
	EvoSkill	63.61 ± 1.92	80.95 ± 3.56	36.94 ± 1.92	43.33 ± 4.86
	SkillGrad (Ours)	**69.44** ± 0.48	**83.34** ± 0.67	**45.83** ± 3.66	**53.81** ± 2.43

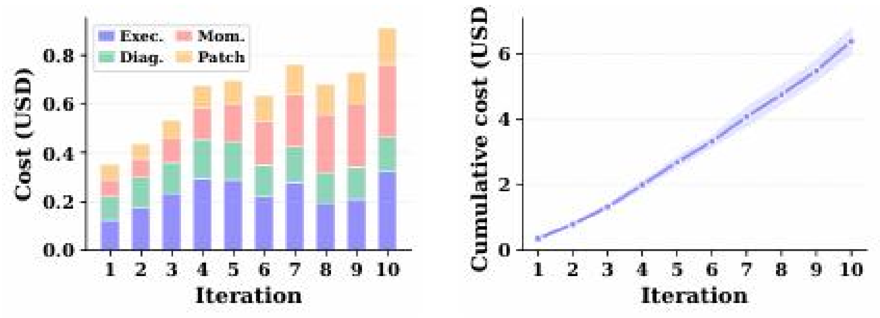
Experiments

消融实验：去除momentum，去除contrastive diagnoses后，出现不同程度性能下降

Variant	Acc.	Δ Acc.
Full SkillGrad	**72.50**	–
No momentum	65.83	−6.67
Failure-only diagnosis	68.33	−4.17



(a) Batch size (b) Training iteration



(a) Per-iteration cost (b) Cumulative cost

其他超参数的影响

Batchsize：更新频率和更新证据的平衡

更多的iteration可能造成skill冲突

Training cost

总共花了6.4 USD

Execution保持持平，主要增长来自momentum，patcher

Experiments

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L3 resource usage

Setting	L3 files	L3 reads	Activation
No skill	0	0	0/120 (0.0)
LLM-gen base	0	0	0/120 (0.0)
Third-party base	4	127	113/120 (94.2)
SkillGrad	$1 \sim 3$	267	259/360 (71.9)

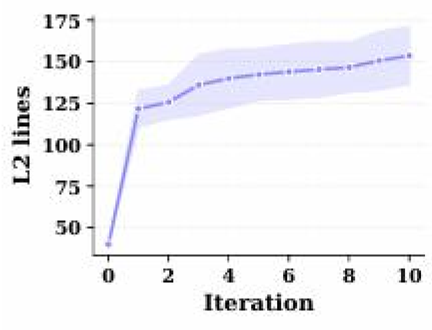
经过skillgrad的优化，L3访问率从0（本来就没有L3）升高到71.9

Third-party base的本来就有l3，但是是general openpyxl references。对比下skillgrad学到的L3更偏procedural and task -conditional

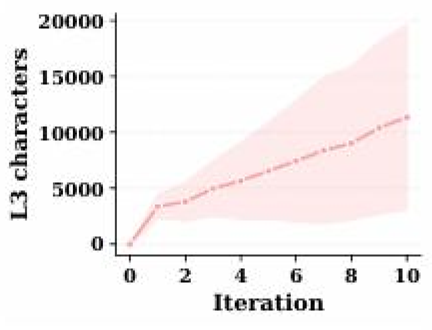
Run	Learned L3 resource	Reads	Behavior stored in L3
seed 0	`mapping_shapes`	40	Structural remaps between source records and destination layouts, including sparse headers, grouped outputs, two-axis lookups, and returned fields from selected records.
seed 0	`formula_vs_python`	38	Decisions between live formulas and Python-written final values, with checks for output domains, formula-produced drivers, repeated paired columns, and post-write verification.
seed 1	`non_target_change_check`	119	Verification that targeted edits do not change unrelated workbook cells.
seed 2	`formula_compatibility`	63	Fallback from fragile formulas to literal values when formulas depend on unsupported functions, dynamic arrays, uncertain recalculation, or mismatched range shapes.
seed 2	`repeated_key_transfers`	6	Duplicate-key transfers, repeated source blocks, and contiguous-run logic without accidental aggregation.
seed 2	`formula_backed_cells`	1	Safe reading of formula-backed helper cells and spilled ranges.

Experiments

Qualitative Training Dynamics



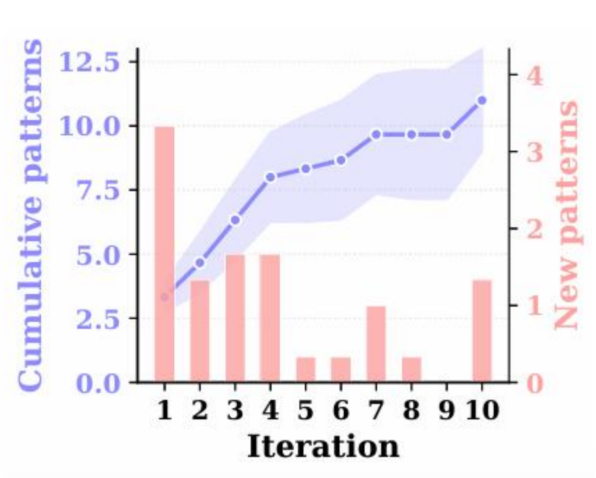
(a) L2 size

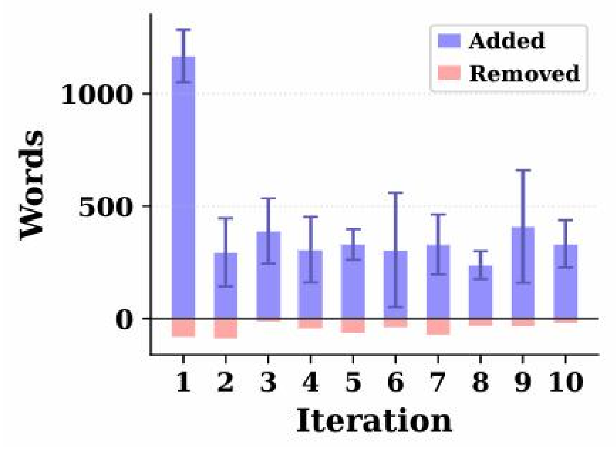


(b) L3 size

L2 第一轮快速增长，后面慢慢refine

L3 单调匀速增长，说明后面新知识被塞进L3（而不是都进SKILL.md）





Cumulative pattern count 在 iteration 7 左右达到10个，然后基本饱和；New patterns 呈现出先高后递减的趋势

Active patterns 从iteration3开始，稳定在4-5个

用Word level衡量更新梯度：第一轮大幅建立结构，后面做较小的增量更新

Experiments

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Qualitative Ablation Analysis

Variant	Acc. (%)	Late train correct	Full batches	L2 words	L3 files/words
SkillGrad	72.50	2.67/4	2	1863	2 / 2786
w/o momentum	65.83	1.83/4	0	2416	3 / 1894

No-momentum run 和 skillgrad 得到的skill总行数差不多，但是组织方式不同，late training阶段的准确率不一样

Variant	Acc. (%)	Pattern record	L2 words	L3 files/words	Representative L3 topics
SkillGrad	72.50	9 op. / 2 wf.	1863	2 / 2786	formula-vs.-Python decisions; mapping shapes
Failure only	68.33	5 op. / 0 wf.	1765	3 / 1217	formula materialization; formula return ranges; grouped block transfers

Failure-only的diagnosis仍然比较强，但是最终的pattern record只包含operation-character patterns

Full SkillGrad 比 failure-only 多解出 5 个 held-out tasks。作者认为差距来自 contrastive diagnosis 捕捉到的 workflow-level behavior

contrastive diagnosis 的价值在于补充 workflow-level 的可复用经验

SkillOpt: Executive Strategy for Self-Evolving Agent Skills

Yifan Yang，Ziyang Gong，etc.
Microsoft, SJTU, Tongji University, Fudan University

SkillOpt：Motivation

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Motivation：skill来自人工写/LLM一次生成/系统基于失败经验随意修改

不像深度学习的优化器一样稳定可控，容易改完反而效果下降

skill 文档本身就是 frozen agent 的外部状态，可以只训练一份skill文档来进行文本空间优化

模型参数对应 skill document，梯度方向对应从轨迹中总结出的 edit direction，学习率对应 edit budget，验证集检查对应 held-out selection gate

Method

Problem setup： D_{tr}， D_{sel}， D_{test}

$$s^\star_{\mathrm{sel}} = \arg\max_{s\in\mathcal{C}(D_{\mathrm{tr}})} \frac{1}{|D_{\mathrm{sel}}|}\sum_{x\in D_{\mathrm{sel}}} r(s),$$

$$(\tau(s), r(s)) = h(M, x, s), \qquad r(s) \in [0,1].$$

$$\mathrm{Test}(s^\star_{\mathrm{sel}}) = \frac{1}{|D_{\mathrm{test}}|}\sum_{x\in D_{\mathrm{test}}} r(s^\star_{\mathrm{sel}}).$$

1. **Forward pass**：收集最终答案、对话、工具调用、verifier反馈和任务特定上下文

2. **Backward pass**：成功和失败轨迹分别划分reflection minibatch，产出结构化编辑建议

3. **Bounded text update**：类似于学习率，每步最多应用L_t个编辑（optimizer model排序）

- 编辑默认是patch mode，也就是局部的append/insert/replace/delete

4. **Validation Gate**：每个 candidate skill 都要在 D_{sel}上重新跑同一个冻结模型和同一个 harness。只有当 selection score 严格高于当前 skill 时，候选 skill 才会被接受。否则这种尝试过的编辑会放进 rejected-edit buffer，后面给optimizer model用。

5. **Epoch-wise slow/meta update**：每个epoch结束后，比较使用上一个skill和当前skill在相同训练样本上运行的结果并总结为improvements/regressions/persistent failures/stable success，更新optimizer-side skill

Experiments

六个benchmark，七个目标模型，三种harness

Model	Skill source	SearchQA	Spreadsheet	OfficeQA	DocVQA	LiveMath	ALFWorld
No harness / direct chat							
GPT-5.5	No skill	77.7	41.8	33.1	78.8	37.6	83.6
	Human skill	81.8$_{+4.1}$	72.9$_{+31.1}$	66.9$_{+33.8}$	90.1$_{+11.3}$	38.4$_{+0.8}$	91.8$_{+8.2}$
	LLM skill	80.9$_{+3.2}$	43.2$_{+1.4}$	51.7$_{+18.6}$	89.6$_{+10.8}$	40.0$_{+2.4}$	93.3$_{+9.7}$
	Trace2Skill	82.4$_{+4.7}$	49.6$_{+7.8}$	65.7$_{+32.6}$	90.6$_{+11.8}$	52.0$_{+14.4}$	87.3$_{+3.7}$
	TextGrad	81.4$_{+3.7}$	41.1$_{-0.7}$	42.0$_{+8.9}$	87.2$_{+8.4}$	49.2$_{+11.6}$	82.8$_{-0.8}$
	GEPA	84.8$_{+7.1}$	73.6$_{+31.8}$	63.9$_{+30.8}$	89.1$_{+10.3}$	43.2$_{+5.6}$	85.8$_{+2.2}$
	SkillOpt	**87.3**$_{+9.6}$	**80.7**$_{+38.9}$	**72.1**$_{+39.0}$	**91.2**$_{+12.4}$	**66.9**$_{+29.3}$	**95.5**$_{+11.9}$
GPT-5.4	No skill	76.9	41.4	50.0	77.6	36.8	75.4
	Human skill	80.1$_{+3.2}$	59.3$_{+17.9}$	59.3$_{+9.3}$	88.5$_{+10.9}$	41.6$_{+4.8}$	74.6$_{-0.8}$
	LLM skill	79.4$_{+2.5}$	46.1$_{+4.7}$	20.4$_{-29.6}$	90.4$_{+12.8}$	36.8$_{+0.0}$	83.6$_{+8.2}$
	Trace2Skill	81.2$_{+4.3}$	53.2$_{+11.8}$	51.2$_{+1.2}$	89.3$_{+11.7}$	40.0$_{+3.2}$	73.1$_{-2.3}$
	TextGrad	82.5$_{+5.6}$	38.6$_{-2.8}$	54.7$_{+4.7}$	85.3$_{+7.7}$	36.6$_{-0.2}$	78.4$_{+3.0}$
	GEPA	82.4$_{+5.5}$	61.1$_{+19.7}$	60.4$_{+10.4}$	88.3$_{+10.7}$	41.6$_{+4.8}$	74.6$_{-0.8}$
	SkillOpt	**83.1**$_{+6.2}$	**62.5**$_{+21.1}$	**62.8**$_{+12.8}$	**91.2**$_{+13.6}$	**44.0**$_{+7.2}$	**91.0**$_{+15.6}$
GPT-5.4-mini	No skill	75.9	36.1	22.1	71.4	14.7	73.1
	Human skill	77.2$_{+1.3}$	42.9$_{+6.8}$	45.9$_{+23.8}$	85.0$_{+13.6}$	28.8$_{+14.1}$	56.7$_{-16.4}$
	LLM skill	76.1$_{+0.2}$	36.8$_{+0.7}$	36.6$_{+14.5}$	86.4$_{+15.0}$	28.0$_{+13.3}$	65.7$_{-7.4}$
	Trace2Skill	78.6$_{+2.7}$	40.7$_{+4.6}$	20.9$_{-1.2}$	88.5$_{+17.1}$	**32.8**$_{+18.1}$	82.8$_{+9.7}$
	TextGrad	77.5$_{+1.6}$	38.2$_{+2.1}$	30.0$_{+7.9}$	84.0$_{+12.6}$	27.2$_{+12.5}$	70.9$_{-2.2}$
	GEPA	79.4$_{+3.5}$	42.5$_{+6.4}$	45.3$_{+23.2}$	83.7$_{+12.3}$	27.2$_{+12.5}$	81.3$_{+8.2}$
	SkillOpt	**80.2**$_{+4.3}$	**47.5**$_{+11.4}$	**48.8**$_{+26.7}$	**90.9**$_{+19.5}$	**32.8**$_{+18.1}$	**85.8**$_{+12.7}$
	No skill	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Experiments

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六个benchmark，七个目标模型，三种harness

在每个模型 x benchmark x harness组合上都达到最优表现

在claude code和codex这种工具执行环境均有效

Codex harness							
	No skill	81.8	27.5	38.3	87.2	35.2	–
	Human skill	84.1 +2.3	50.7 +23.2	40.0 +1.7	88.8 +1.6	48.8 +13.6	–
GPT–5.5	LLM skill	83.4 +1.6	25.0 −2.5	34.3 −4.0	89.8 +2.6	45.6 +10.4	–
	EvoSkill	61.4 −20.4	67.5 +40.0	42.4 +4.1	89.3 +2.1	63.2 +28.0	–
	SkillOpt	**87.3** +5.5	**85.0** +57.5	**51.1** +12.8	**92.2** +5.0	**78.4** +43.2	–
Claude Code harness							
	No skill	81.9	22.1	57.6	86.6	40.8	–
	Human skill	83.7 +1.8	37.1 +15.0	66.3 +8.7	88.0 +1.4	44.0 +3.2	–
GPT–5.5	LLM skill	82.4 +0.5	37.1 +15.0	56.4 −1.2	89.6 +3.0	44.8 +4.0	–
	EvoSkill	84.0 +2.1	75.0 +52.9	70.3 +12.7	87.2 +0.6	52.0 +11.2	–
	SkillOpt	**85.9** +4.0	**80.4** +58.3	**71.5** +13.9	**90.1** +3.5	**56.5** +15.7	–

Experiments

三个关键模块的设计是有用的；超参数不敏感

Component	Setting	SearchQA	SpreadsheetBench	LiveMath
Learning-rate form	lr=4 (default)	**87.1**	**77.5**	**61.3**
	dynamic lr	85.8	71.8	54.0
	without lr	84.6	75.7	57.3
Rejected buffer	with rejected buffer	**87.1**	**77.5**	**61.3**
	without rejected buffer	85.5	72.9	58.9
Slow/meta update	meta skill and slow update	**87.1**	**77.5**	**61.3**
	without meta skill	85.1	75.7	58.1
	without meta skill and slow update	86.3	55.0	59.7

(a) Training set size

Setting	SearchQA	Spreadsheet	LiveMath
1 example	81.0	47.5	59.1
20% train	84.1	69.0	65.9
40% train	**86.1**	73.5	64.8
80% train	**86.1**	77.6	67.0
100% train	84.1	**78.0**	**70.5**

(b) Mini-batchsize

Setting	SearchQA	Spreadsheet	LiveMath
1	85.9	75.4	60.5
2	86.3	77.1	54.8
4	86.9	75.4	**64.5**
8	**87.1**	77.5	61.3
16	87.0	**77.9**	61.3
32	86.9	77.5	58.9

(c) Batchsize

Setting	SearchQA	Spreadsheet	LiveMath
8	85.1	76.8	58.1
24	86.4	77.1	**62.9**
40	87.1	**77.5**	61.3
56	86.5	76.8	56.5
full epoch	**87.2**	75.0	53.2

(d) Learning rate

Setting	SearchQA	Spreadsheet	LiveMath
lr=1	85.5	77.5	62.1
lr=2	86.7	77.5	60.5
lr=4	86.5	**78.2**	56.5
lr=8	**87.0**	73.6	**66.9**
lr=16	86.8	**78.2**	65.3

(e) Learning-rate scheduler

Setting	SearchQA	Spreadsheet	LiveMath
constant	**87.3**	**80.7**	62.1
cosine	87.1	77.5	61.3
linear	87.2	72.9	**62.9**

(f) Slow-update samples

Setting	SearchQA	Spreadsheet	LiveMath
5	86.8	76.4	64.5
10	86.4	74.3	**65.3**
20	**87.1**	**77.5**	61.3
40	86.9	75.4	54.8

Experiments

跨模型、跨harness、跨benchmark迁移均显示出正提升

Benchmark：share only the broad task family (math)

(a) Cross-model transfer					
Source model	**Target model**	**Benchmark**	**Baseline**	**Direct**	**Transferred**
GPT–5.4	GPT–5.4	SpreadsheetBench	41.4	62.5	–
	GPT–5.4-mini		36.1	47.5	$45.5_{+9.4}$
	GPT–5.4-nano		23.5	42.5	$26.5_{+3.0}$
GPT–5.4	GPT–5.4	LiveMath	36.8	44.0	–
	GPT–5.4-mini		14.7	32.8	$19.2_{+4.5}$
	GPT–5.4-nano		23.2	27.2	$28.8_{+5.6}$
(b) Cross-harness transfer					
Source harness	**Target harness**	**Benchmark**	**Baseline**	**Direct**	**Transferred**
Codex	Claude Code	LiveMath	40.8	56.5	$42.4_{+1.6}$
Claude Code	Codex		35.2	78.4	$48.0_{+12.8}$
Codex	Claude Code	SpreadsheetBench	22.1	80.4	$81.8_{+59.7}$
Claude Code	Codex		27.5	85.0	$71.1_{+43.6}$
(c) Cross-benchmark transfer					
Source benchmark	**Target benchmark**	**Model**	**Baseline**	**Direct**	**Transferred**
OlympiadBench	Omni-MATH	GPT–5.4	56.6	–	$60.3_{+3.7}$
		GPT–5.4-mini	34.8	–	$36.6_{+1.8}$
		GPT–5.4-nano	38.8	–	$40.1_{+1.3}$

Experiments

更强的模型作为optimizer效果更好（优于使用target matched模型）

和蒸馏的方法不同，只使用target model本身作为optimizer也可以获得56%-70%的性能提升

Benchmark	Target	Baseline	Strong optimizer (GPT–5.5)	Target-matched optimizer
SpreadsheetBench	GPT–5.4-mini	36.1	**47.5** +11.4	43.2 +7.1
	GPT–5.4-nano	23.5	**42.5** +19.0	35.4 +11.9
SearchQA	GPT–5.4-mini	75.9	**80.2** +4.3	78.3 +2.4
	GPT–5.4-nano	55.8	**74.8** +19.0	69.9 +14.1

Experiments：Compactness, Cost, Examples

- 最终的token从379到1995之间

“Well below a typical system-prompt budget for modern frontier models”

- Validation是有用的：非常少的edit就带来了很大性能提升（+39， +26）
- SkillOpt 离线训练每换来 1 个测试集百分点，需要花的训练token数在0.6M到46.4M

Benchmark	Initial (tok)	Final (tok)	Edits	Train tokens	Cost / pt
SearchQA	16	857	4	213.8M	37.9M
SpreadsheetBench	224	1,995	4	21.4M	0.6M
OfficeQA	145	883	1	20.8M	1.1M
DocVQA	81	959	3	188.2M	46.4M
LiveMath	154	379	1	23.2M	3.6M
ALFWorld	516	1,321	2	59.3M	15.9M

Experiments：Compactness, Cost, Examples

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每个benchmark分别摘出代表性规则进行分析：

- 规则是procedural，而不是instance-specific
- 规则补的是 frontier model 零样本时容易缺失的：answer-format constraints，evidence binding to a specific visual region，workbook-structure-first reasoning等
- 像人类专家试用 benchmark 一天后总结出来的经验

SearchQA. "Infer the expected answer type from clue wording, then choose the shortest canonical entity supported by co-occurring distinctive evidence."

SpreadsheetBench. "Inspect workbook structure and formulas, then write evaluated static values across the full requested target range instead of relying on Excel recalculation."

OfficeQA. "Treat oracle parsed pages as primary evidence, lock table/date/unit context, and output exactly the requested rounded value without extra labels."

DocVQA. "For tables, forms, charts, and legends, first bind the question to the exact visual row/header/field, then copy only the aligned answer span."

LiveMathematicianBench. "In strongest-statement MCQs, rank choices by theorem strength and prefer a justified stronger-result option over true but weaker corollaries."

ALFWorld. "Keep a horizon-aware visited/frontier ledger, diversify search after repeated same-type failures, and avoid revisiting the destination until holding the target."